II. Volume of both shots

$$= \frac{4}{3}\pi(0.75)^3 + \frac{4}{3}\pi(0.8)^3$$

$$= \frac{4}{3}\pi\left[\left(\frac{3}{4}\right)^3 + \left(\frac{4}{5}\right)^3\right]$$

$$= \frac{4}{3}\pi\left(\frac{27}{64} + \frac{64}{125}\right)$$

$$= \frac{4}{3}\pi\left(\frac{3375 + 4096}{8000}\right)$$

$$= \frac{4}{3}\pi\left(\frac{7471}{8000}\right) = \frac{4}{3}\pi(0.93)\text{ cm}^3$$

Hence, statement I, II are true.

70. *(c)* I. Coterminous edges are those who have same boundaries and for a cuboid, these may be considered as length, breadth and height.

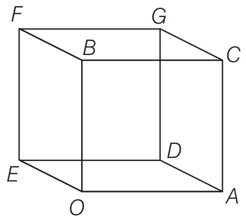
So, given statement is true

II. The surface area of a cuboid is twice the sum of the products of lengths of its coterminous edges taken two at a time.

Hence, both statements I and II are correct.

71. *(d)* I. The distance between vertices B and C is 1 cm.

II. The distance between A and B is $\sqrt{1^2 + 1^2} = \sqrt{2}$ cm



III. The distance between vertices B and D is $\sqrt{1^2 + 1^2 + 1^2} = \sqrt{3}$ cm

Hence, the statements I and II, III are correct.

72. *(d)* I. It is true that the surface area of all the eight drops is greater than the surface area of big drop.

II. Let radius of spherical body be 'a'.

$\therefore$ Volume of sphere, $V = \frac{4}{3}\pi a^3$

$\therefore$ Surface area of sphere, $S = 4\pi a^2$

$$\Rightarrow \quad V^2 = \frac{16}{9}\pi^2 a^6 \propto S^3$$

III. It is true that sphere occupies biggest space but has smallest surface area.

Hence, all statements are true.

73. *(c)* I. It is a correct statement.

II. The length may be πr but the breadth is always less than $\sqrt{h^2 + r^2}$ of formed rectangle,

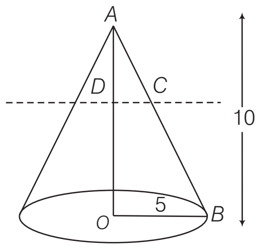
So, statement II is false.

Statement III and IV are correct.

Solution. (74-76)

As cone is divided into two parts by a plane through mid-point of its axis, therefore it will bisect the slant height.

$\Delta AOB \sim \Delta ADC$



$$\therefore \frac{DC}{OB} = \frac{AD}{AO} = \frac{1}{2} \Rightarrow DC = \frac{OB}{2}$$

$\therefore$ Radius of smaller cone will be half of original cone

74. *(c)* The volume of the original cone

$$= \frac{1}{3} \times \pi \times (5)^2 \times 10 = \frac{250\pi}{3}$$

The volume of the frustum.

$$= \frac{250\pi}{3} - \frac{1}{3} \times \pi\left(\frac{5}{2}\right)^2 \times 5$$

$$= \frac{250\pi}{3} - \frac{125\pi}{12} = \frac{875\pi}{12}$$

Thus, dividing the two volumes, we get $\frac{8}{7}$

75. *(c)* Area of the top circle of the frustum

$$= \pi\left(\frac{5}{2}\right)^2 = \frac{25\pi}{4}$$

76. *(d)* Let l be slant height of original cone.

Curved surface area of cone

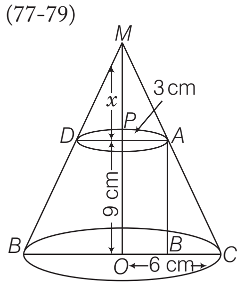
$= \pi \times 5 \times l = 5\pi l$

Curved surface area of frustum

$$= 5\pi l - \pi \times \frac{5}{2} \times \frac{l}{2} = \frac{15}{4}\pi l$$

Thus, the ratio of the two will be $\frac{4}{3}$ or 4 : 3.

Solution. (77-79)



Here, $r_1 = 3$ cm, $r_2 = 6$ cm and $h = 9$ cm

77. *(a)* In ΔABC, $AC^2 = AB^2 + BC^2$

$AC^2 = 9^2 + (6-3)^2 = 81 + 9 = 90$

$\therefore AC = 3\sqrt{10}$ cm

78. *(d)* By using properties of similar triangle in ΔMPA and ΔMOC.

$$\frac{3}{6} = \frac{x}{x+9} \Rightarrow 3x + 27 = 6x$$

$\Rightarrow \quad 27 = 3x$

$\Rightarrow \quad x = 9$ cm

Hence, height of cone = 9 + 9

= 18 cm

79. *(b)* Total surface area of the frustum

$= \pi[(R + r)l + r^2 + R^2]$

where, $l = \sqrt{h^2 + (R-r)^2}$

$= \pi[(6+3)\sqrt{81+9} + 9 + 36]$

$= \pi(9\sqrt{90} + 45)$

$= 9\pi(3\sqrt{10} + 5)$ cm^2

80. *(c)* Curved surface area of the well

$$= 2\pi rh = 2 \times \frac{22}{7} \times 2 \times 14 = 176 \text{ m}^2$$

$\because$ Expense of getting per square metre plastered = ₹ 25

$\therefore$ Expense of 176 m^2 = 176×25

= ₹ 4400

81. *(a)* Given that,

Radius of sphere (r) = 9 cm = 0.09 m

and diameter of wire (d) = 0.4 cm

$\Rightarrow R = 0.2$ cm = 0.002 m

Given condition,

Volume of sphere = Volume of wire

$$\Rightarrow \frac{4}{3}\pi r^3 = \pi R^2 h \Rightarrow h = \frac{4}{3} \cdot \frac{r^3}{R^2}$$

$$= \frac{4}{3} \times \frac{0.09 \times 0.09 \times 0.09}{0.002 \times 0.002} = 81 \times = 243 \text{ m}$$

82. *(c)* Total surface area of cube

$= 6 \times (\text{Side})^2$

$\therefore \quad 150 = 6 \times (\text{Side})^2$

$$\Rightarrow \quad \text{Side}^2 = \frac{150}{6} = 25$$

$\therefore \quad$ Side = $\sqrt{25}$ = 5 cm

$\therefore$ Volume of cube = (Side)^3

$= 5 \times 5 \times 5 = 125$ cm^3

83. *(b)* Volume of cube = (Side)^3

$\because \quad 729 = a^3 \Rightarrow a = 9$ cm

$\therefore$ Diagonal of cube = Side $\times \sqrt{3}$

$= 9 \times \sqrt{3} = 9\sqrt{3}$ cm

84. *(a)* Curved surface area of right circular cone = $\pi r l$

$$\therefore 440 = \frac{22}{7} \times 14 \times l \Rightarrow l = \frac{440 \times 7}{22 \times 14}$$

= 10 cm

85. *(d)* Volume of cylinder $= \pi r_1^2 h$

Volume of sphere $= \frac{4}{3}\pi r_2^3$

Number of spheres = 48

$$\therefore \frac{\text{Volume of cylinder}}{\text{Volume of sphere}} = \frac{\pi r_1^2 h}{\frac{4}{3}\pi r_2^3}$$

$$\Rightarrow \frac{\pi r_1^2 h}{\frac{4}{3}\pi r_2^3} = 48 \Rightarrow \frac{\pi r_1^3}{\frac{4}{3}\pi r_2^3} = 48 \quad [\because r_1 = h]$$

$$\Rightarrow \frac{3}{4}\left(\frac{r_1}{r_2}\right)^3 = 48 \Rightarrow \left(\frac{r_1}{r_2}\right)^3 = \frac{48 \times 4}{3}$$

$$\Rightarrow \frac{r_1}{r_2} = (64)^{1/3} \Rightarrow \frac{r_1}{r_2} = \frac{4}{1} \Rightarrow \frac{r_2}{r_1} = \frac{1}{4}$$

Hence, the ratio of radius of ball to cylinder is 1 : 4.

86. *(a)* Volume of the cuboid = 720 cm^3

Area of base $= lb = 72$

Height of the cuboid

$$= \frac{\text{Volume of the cuboid}}{\text{Base area of the cuboid}}$$

$$= \frac{720}{72} = 10 \text{ cm}$$

Surface area of the cuboid

$$= 2(lb + bh + hl)$$

$$\Rightarrow 484 = 2(72 + b \times 10 + l \times 10)$$

$$\Rightarrow 20(l + b) = 340 \Rightarrow l + b = 17$$

So, it is obvious that length, breadth and height of the cuboid is 9 cm, 8 cm and 10 cm.

87. *(a)* Surface area of the sphere

$$= 4\pi r^2$$

$$\Rightarrow 616 = 4\pi r^2 \Rightarrow \pi r^2 = \frac{616}{4} = 154$$

$$\Rightarrow r^2 = \frac{154 \times 7}{22} = 49$$

$$\because r = \sqrt{49} = 7 \text{ cm}$$

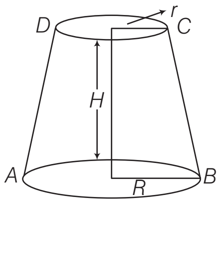
$$\therefore \text{Volume of the sphere} = \frac{4}{3}\pi r^3$$

$$= \frac{4}{3} \times \frac{22}{7} \times 7 \times 7 \times 7 = \frac{4312}{3} \text{ cm}^3$$

88. *(a)* Given that,

Area of first end $= P = \pi r^2$

and area of second end $= Q = \pi R^2$



Given, $P < Q \Rightarrow r = \sqrt{\frac{P}{\pi}}$ and $R = \sqrt{\frac{Q}{\pi}}$

$\therefore$ Difference in radii of the ends of the frustum $= R - r$

$$= \sqrt{\frac{Q}{\pi}} - \sqrt{\frac{P}{\pi}} = \frac{\sqrt{Q} - \sqrt{P}}{\sqrt{\pi}}$$

89. *(c)* We know that,

Volume of frustum

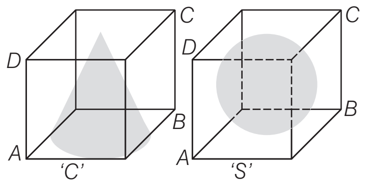
$$= \frac{\pi H}{3}(R^2 + r^2 + Rr)$$

$$= \frac{\pi}{3} H \left\{\left(\sqrt{\frac{Q}{\pi}}\right)^2 + \left(\sqrt{\frac{P}{\pi}}\right)^2 + \sqrt{\frac{Q}{\pi}} \cdot \sqrt{\frac{P}{\pi}}\right\}$$

$$= \frac{\pi H}{3}\left\{\frac{Q}{\pi} + \frac{P}{\pi} + \frac{\sqrt{PQ}}{\pi}\right\}$$

$$= \frac{H}{3}(P + Q + \sqrt{PQ})$$

90. *(b)* Let the side of both cubes be a, then the height and radius of a cone, $h = a$ and $r = \frac{a}{2}$



and radius of sphere $(R) = \frac{a}{2}$

$$\therefore \text{Volume of cone } (C) = \frac{1}{3}\pi r^2 h$$

$$= \frac{1}{3}\pi\left(\frac{a}{2}\right)^2 a \Rightarrow C = \frac{\pi a^3}{12} \quad \text{...(i)}$$

and volume of sphere $(S) = \frac{4}{3}\pi R^3$

$$= \frac{4}{3}\pi\left(\frac{a}{2}\right)^3 = \frac{\pi a^3}{6} \quad \text{... (ii)}$$

From Eqs. (i) and (ii), $S = 2C$

91. *(d)* If 10 circular plates each of thickness 3 cm, are placed one above the other, then it forms a cylinder with height ($3 \times 10 = 30$ cm) and a hemisphere of radius 6 cm is placed on the top just to cover the cylinder that means its radius is also 6 cm.

$\therefore$ Radius of hemi-sphere $(R) = 6$ cm

Radius of cylinder $(r) = 6$ cm

and height of cylinder $(h) = 30$ cm

$\therefore$ Volume of the solid = Volume of cylinder + Volume of hemisphere

$$= \pi r^2 h + \frac{2}{3}\pi R^3$$

$$= \pi(6)^2 \times 30 + \frac{2}{3}\pi(6)^3$$

$$= \pi \times 36 \times 30 + \frac{2}{3}\pi \times 216$$

$$= 1080\pi + 2\pi \times 72$$

$$= 1080\pi + 144\pi = 1224\pi \text{ cm}^3$$

92. *(a)* Given that, the height and radius of a right circular metal cone (solid) are 8 cm and 2 cm, respectively.

i.e. $h = 8$ cm and $r = 2$ cm

Let the radius of the sphere be R.

Then, by condition, $\frac{1}{3}\pi r^2 h = \frac{4}{3}\pi R^3$

$$\Rightarrow 4 \times 8 = 4R^3 \Rightarrow R^3 = (2)^3$$

$$\therefore R = 2$$

Hence, radius of the sphere = 2 cm

93. *(c)* Let the edge of a square be x. Then, its volume is x^3 and sum of its edges is $12x$.

Now, by condition, $x^3 = 12x$

$$\Rightarrow x(x^2 - 12) = 0$$

$$\Rightarrow x^2 = 12 \quad [\because x \neq 0]$$

$$\therefore \text{Its total surface area} = 6x^2$$

$$= 6(12) = 72 \text{ sq units}$$

94. *(a)* Given that,

Diameter of a right circular cone = 7 cm

$\therefore$ Radius of a right circular cone $= \frac{7}{2}$ cm

and slant height of a right circular cone $(l) = 10$ cm

$\therefore$ Lateral surface area of a cone $= \pi r l$

$$= \frac{22}{7} \times \frac{7}{2} \times 10 = 11 \times 10 = 110 \text{ cm}^2$$

95. *(d)* Let the height and radius of solid cylinder be h and r cm, respectively.

Given that, $r = 5$ cm

and total surface area = 660 cm^2

$$\Rightarrow 2\pi rh + 2\pi r^2 = 660$$

$$\Rightarrow 2\pi r(h + r) = 660$$

$$\Rightarrow (h + 5) = \frac{330}{5\pi} = \frac{330}{5} \times \frac{7}{22}$$

$$\Rightarrow h = \frac{66 \times 7}{22} - 5 = 21 - 5 = 16$$

$\therefore$ Required height = 16 cm

96. *(a)* Let the diameters of two spheres be d_1 and d_2, respectively.

$$\because d_1 : d_2 = 3:5$$

$\therefore$ Ratio of their surface areas

$$= \frac{4\pi r_1^2}{4\pi r_2^2} = \frac{(2r_1)^2}{(2r_2)^2} = \frac{d_1^2}{d_2^2}$$

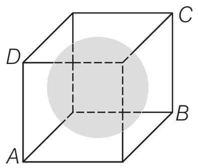
$$= \left(\frac{d_1}{d_2}\right)^2 = \left(\frac{3}{5}\right)^2 = \frac{9}{25} = 9 : 25$$

97. *(c)* From figure, it is clear that

Diameter of a sphere = Side of the cube = 3 cm

$$\because \text{Radius} = \frac{3}{2} \text{ cm}$$

$\therefore$ Volume of the largest sphere



$= \frac{4}{3}\pi\,(\text{Radius})^3 = \frac{4}{3}\pi\left(\frac{3}{2}\right)^3$

$= \frac{4}{3}\pi \cdot \frac{27}{8} = \frac{9}{2}\pi = 4.5\pi\ \text{cm}^3$

98. *(d)* Given that, radius of a right circular cone $(r) = 12$ m

Height of a right circular cone $(h) = 5$ m

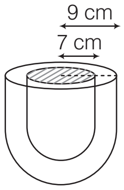
$\therefore$ Lateral height of cone $(l) = \sqrt{r^2 + h^2}$

$= \sqrt{144 + 25} = \sqrt{169} = 13$ m

$\therefore$ Required quantity of cloth to roll up to form a right circular tent $= \pi r l$

$= \pi \times (12) \times (13) = 156\ \pi\ \text{m}^2$

99. *(a)* Given that, Outer radius of hemispherical shell $(R) = 9$ cm



and inner radius of hemispherical shell $(r) = 7$ cm

$\therefore$ Volume of a hemispherical shell

$= \frac{2}{3}\pi\,(R^3 - r^3)$

$= \frac{2}{3} \times \frac{22}{7} \times (729 - 343)$

$= \frac{2}{3} \times \frac{22}{7} \times 386 = \frac{16984}{21} = 808.76$

$\approx 808\ \text{cm}^3$ (approx)

100. *(b)* Let r be the radius of the sphere.

Given that, volume of sphere $= 36\ \pi$

$\Rightarrow \frac{4}{3}\pi r^3 = 36\ \pi \Rightarrow r^3 = 27 = (3)^3$

$\because$ $r = 3$ cm

$\therefore$ Diameter of sphere

$= 2r = 2\,(3) = 6$ cm

and surface area of sphere $= 4\pi r^2$

$= 4\ \pi\,(3)^2 = 36\ \pi\ \text{cm}^2$

$\therefore$ Required ratio

$= \frac{\text{Surface area of sphere}}{\text{Diameter of sphere}} = \frac{36\ \pi}{6} = 6\pi$

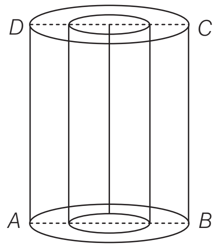
101. *(d)* Given that,

Internal diameter of the tube = 6 cm

$\therefore$ Internal radius $(r) = 3$ cm

Height of the tube $(h) = 10$ cm

Thickness of the metal = 1 cm



$\because$ Outer radius (R)

= Thickness of the metal + Internal radius

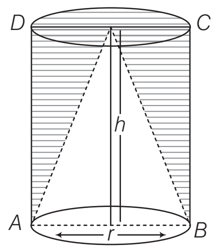
$= 1 + 3 = 4$ cm

$\therefore$ Outer curved surface area

$= 2\pi rh = 2 \times \pi \times 4 \times 10 = 80\pi$

102. *(a)* Let the height and radius of right circular cylinder be h and r, respectively.

Then, volume of cylinder $= \pi r^2 h$



Volume of circular cone $= \frac{1}{3}\pi r^2 h$

$\therefore$ Required ratio

$= \frac{\text{Volume of utilised wood}}{\text{Volume of wasted wood}}$

$= \frac{\text{Volume of right circular cone}}{\text{Volume of right circular cylinder} - \text{Volume of right circular cone}}$

$= \frac{\frac{1}{3}\pi r^2 h}{\pi r^2 h - \frac{1}{3}\pi r^2 h} = \frac{\frac{1}{3}\pi r^2 h}{\frac{2}{3}\pi r^2 h} = \frac{1}{2} = 1:2$

103. *(c)* Since, volume of cone and pyramid $= \frac{1}{3} \times$ Base area $\times$ Height

Therefore, they have same volume but their surface areas are not same as nothing can be said about their slant heights.

104. *(b)* Given that, height of cone $(h) = 3$ cm and slant height of cone $(l) = 5$ cm

Let r be the radius of cone.Then,

$l = \sqrt{r^2 + h^2} = 5 \Rightarrow r^2 + h^2 = 25$

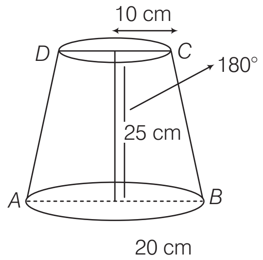
$\Rightarrow r^2 = 25 - 9 = 16 \Rightarrow r = 4$ cm

$\therefore$ Volume of cone $= \frac{1}{3}\pi r^2 h$

$= \frac{1}{3} \times \frac{22}{7} \times 16 \times 3 = \frac{352}{7} = 50.3\ \text{cm}^3$

105. *(a)* Given that,

Height of bucket = 25 cm



Radius of top $(R) = 20$ cm

and radius of bottom $(r) = 10$ cm

$\therefore$ Capacity of bucket

$= \frac{\pi}{3} h\,(R^2 + r^2 + rh)$

$= \frac{\pi}{3} \times 25\,(400 + 100 + 200)\ \text{cm}^3$

$= \frac{\pi}{3} \times 25 \times 700\ \text{cm}^3$

$= \frac{\pi}{3} \times \frac{175 \times 100}{1000}\ \text{L} = \frac{17.5\pi}{3}\ \text{L}$

106. *(d)* Given that, $h = 15$ cm

Let r be the radius of cylinder.

Lateral surface area of cylinder $= 2\pi rh$

$\Rightarrow 2\pi rh = 660$ [given]

$\Rightarrow \pi rh = 330$

$\Rightarrow \frac{22}{7} \times r \times 15 = 330 \Rightarrow \frac{22}{7} \times r = 22$

$\because$ $r = 7$ cm

$\therefore$ Volume of cylinder $= \pi r^2 h$

$= \frac{22}{7} \times 49 \times 15$

$= 22 \times 7 \times 15 = 2310\ \text{cm}^3$

107. *(b)* Given that, the diameter of Moon is approximately one-fourth of the diameter of Earth.

If radius of Moon $= r$

Then, radius of Earth $= 4r$

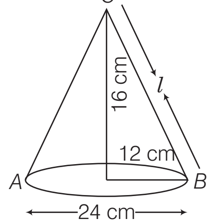
$\therefore \frac{\text{Volume of Moon}}{\text{Volume of Earth}} = \frac{\frac{4}{3}\pi r^3}{\frac{4}{3}\pi\,(4r)^3}$

$= \frac{r^3}{64\,r^3} = \frac{1}{64} = 1:64$

108. *(c)* Slant height, $l = \sqrt{h^2 + r^2}$

$= \sqrt{16^2 + 12^2}$

$= \sqrt{256 + 144} = \sqrt{400} = 20$ cm



Curved surface area $= \pi rl$

$$= \frac{22}{7} \times 12 \times 20 \text{ cm}^2$$

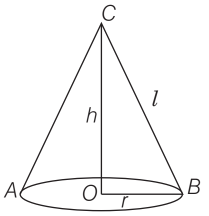
Cost of painting the surface of the cap

$$= \frac{22}{7} \times 12 \times 20 \times 0.70 = ₹\ 528$$

109. *(b)* Slant height, $l = \sqrt{h^2 + r^2}$

$$= \sqrt{(24)^2 + (7)^2} = \sqrt{576 + 49}$$

$$= \sqrt{625} = 25$$



Total surface area $= \pi r\,(l + r)$

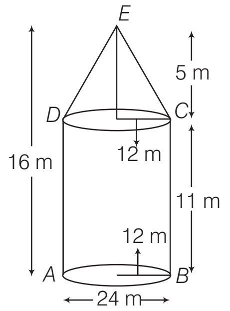
$$= \frac{22}{7} \times 7\,(25 + 7)$$

$$= \frac{22}{7} \times 7 \times 32 = 704 \text{ cm}^2$$

110. *(c)* Radius of cone, $r = \frac{24}{2} = 12$ cm

Slant height of the cone, $l = \sqrt{5^2 + 12^2}$

$$= \sqrt{25 + 144} = \sqrt{169} = 13 \text{ m}$$



Curved surface area for conical portion

$$= \pi rl = \frac{22}{7} \times 12 \times 13 = \frac{3432}{7} \text{ m}^2$$

111. *(d)* Curved surface area of cylinder

$= 2\pi rh = x$

Volume of cylinder $= \pi r^2 h = y$

$$\Rightarrow \quad \frac{2\pi rh}{\pi r^2 h} = \frac{x}{y} \Rightarrow r = \frac{2y}{x}$$

Also, $\quad h = \frac{x}{2\pi r}$

$$\therefore \quad \text{Required ratio} = \frac{h}{r} = \frac{\frac{x}{2\pi r}}{\frac{2y}{x}}$$

$$= \frac{x}{2\pi \cdot \frac{2y}{x}} \times \frac{x}{2y} = \frac{x^3}{8\pi y^2}$$

So, the ratio is not independent of x or y.

112. *(a)* Let common radius be r cm.

Then, height of cylinder $= h$

and height of cone $= h'$

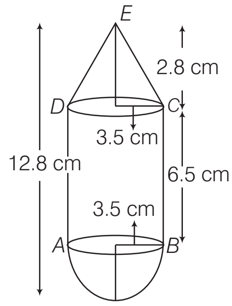
$\therefore$ Volume of the complete structure

$$= \frac{1}{3}\pi r^2 h' + \pi r^2 h + \frac{2}{3}\pi r^3$$

$$= \pi r^2 \left(\frac{h'}{3} + h + \frac{2}{3} r\right)$$

$$= \pi\,(3.5)^2 \left(\frac{2.8}{3} + 6.5 + \frac{2}{3} \times 3.5\right)$$

$$= \pi \times 3.5 \times 3.5 \times 9.76 = 375.86 \text{ cm}^3$$



Hence, volume (V) of the structure lies between 370 cm^3 and 380 cm^3.

113. *(b)* Given, radius of cone $= \frac{6}{2} = 3$ cm

and height of cone = 4 cm

Now, curved surface area $= \pi rl$

where, $l = \sqrt{r^2 + h^2} = \sqrt{3^2 + 4^2} = 5$ cm

$\therefore$ Curved surface area $= \pi \times 3 \times 5 = 15\pi$

$$= \frac{15 \times 22}{7} \approx 47 \text{ cm}^2$$

114. *(d)* Volume of clay required

$$= \pi\left[\left(\frac{5.1}{2}\right)^2 - \left(\frac{4.5}{2}\right)^2\right] \times 21$$

$$= \pi\,[(2.55)^2 - (2.25)^2] \times 21$$

$$= \pi\,(0.3 \times 4.8) \times 21 = 30.24\,\pi \text{ cm}^2$$

115. *(a)* Surface area of cube which can be painted $= 6\,(\text{Side})^2 = 6(2)^2 = 24 \text{ cm}^2$

Now, surface area of cuboid which can be painted

$$= 2(lb + bh + lh)$$

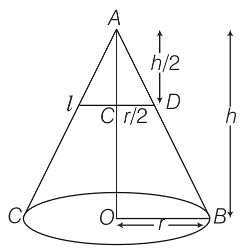
$$= 2(2 + 6 + 3) = 22 \text{ cm}^2$$

Total surface area of both cube and cuboid

$$= 22 + 24 = 46 \text{ cm}^2 < 54 \text{ cm}^2$$

Hence, both cube and cuboid can be painted.

116. *(b)* Let the cone is divided into two parts by a line l.



In ΔAOB and ΔACD, $\Delta AOB \sim \Delta ACD$

By basic proportionality theorem,

$$CD = \frac{r}{2}, \text{ since } AC = \frac{h}{2}$$

$$\text{Now, ratio} = \frac{\text{Volume of original cone}}{\text{Volume of smaller cone}}$$

$$= \frac{\frac{1}{3}\pi r^2 h}{\frac{1}{3}\pi\left(\frac{r}{2}\right)^2\left(\frac{h}{2}\right)} = \frac{8}{1}$$

$\therefore$ Required ratio = 8 : 1

117. *(c)* Volume of each small sphere

$$= \frac{\text{Volume of bigger sphere}}{\text{Number of small spheres}} = \frac{\frac{4}{3}\pi(4)^3}{64}$$

$$= \frac{4}{3} \times \frac{\pi \times 4 \times 4 \times 4}{64} = \frac{4}{3}\pi \text{ cm}^3$$

Let radius of small sphere be r'.

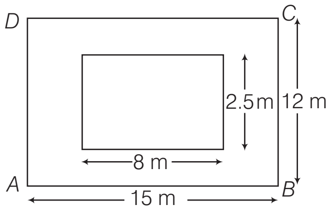
$$\therefore \quad \frac{4}{3}\pi r'^3 = \frac{4}{3}\pi \quad \Rightarrow r' = 1 \text{ cm}$$

Now, surface area of small sphere

$$= 4\pi r'^2 = 4\pi \text{ cm}^2$$

118. *(c)* Volume of Earth dug out

$$= 8 \times 2.5 \times 2 = 40 \text{ m}^3$$



Area where Earth is spread × Field level raised

= Volume of Earth dug out

$$\Rightarrow \quad [(12 \times 15) - (8 \times 2.5)] \times h = 40$$

$$\therefore \quad h = \frac{40}{180 - 20} = \frac{40}{160} = \frac{1}{4} \text{ m}$$

$$= \frac{100}{4} \text{ cm} = 25 \text{ cm}$$

119. *(a)* Given, surface area of sphere

$$= 616 \text{ cm}^2$$

$$\Rightarrow \quad 4\pi r^2 = 616 \Rightarrow r^2 = \frac{616 \times 7}{4 \times 22}$$

$$\Rightarrow \quad r^2 = 7 \times 7 \quad \therefore \quad r = 7 \text{ cm}$$

$\therefore$ Diameter of the largest circle lying on sphere

$$= 2 \times r = 14 \text{ cm}$$

120. *(c)* Given, volume of cube $= 216x^3$

$(\text{Side})^3 = 216x^3 \Rightarrow$ Side $= 6x$

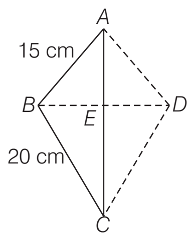
Since, sphere is enclosed in hollow cube.

$\therefore$ Diameter of sphere $= 6x$

Now, surface area of sphere $= 4\pi r^2$

$$= 4\pi\left(\frac{6x}{2}\right)^2 = 36\,\pi x^2$$

121. *(c)* Let ABC be a right angled triangle.
Then, hypotenuse, $AC = 25$ cm



Let $AB = 3x$ and $BC = 4x$

Thus, $AC^2 = AB^2 + BC^2$ [by pythagoras theorem]

$\Rightarrow \quad (25)^2 = (3x)^2 + (4x)^2$

$\Rightarrow \quad (25)^2 = 9x^2 + 16x^2$

$\Rightarrow \quad 25 = x^2 \Rightarrow x = 5$

$\therefore \quad AB = 15$ cm and $BC = 20$ cm

Now, ΔABC revolves about AC, so it forms two cones ABD and BCD.

Since, ΔAEB and ΔABC are similar.

$\therefore \quad \frac{BE}{BC} = \frac{AB}{AC} \Rightarrow \frac{BE}{20} = \frac{15}{25}$

$\Rightarrow \quad BE = \frac{15 \times 20}{25} = 12$ cm

So, radius of the base of cone

$= BE = 12$ cm

In right angled ΔAEB,

$AE = \sqrt{(AB)^2 - (BE)^2}$

$= \sqrt{(15)^2 - (12)^2}$

$= \sqrt{225 - 144} = \sqrt{81} = 9$ cm

So, height of cone $ABD = AE = 9$ cm

$\therefore$ Height of cone $BCD = AC - AE$

$= 25 - 9 = 16$ cm

Now, volume of cone $ABD = \frac{1}{3}\pi r^2 h$

$= \frac{1}{3}\pi (12)^2 \times 9 = 432\pi \text{ cm}^3$

and volume of cone

$BCD = \frac{1}{3}\pi (12)^2 \times 16 = 768\pi \text{ cm}^3$

$\therefore$ Required volume of double cone

$= 432\pi + 768\pi = 1200\pi$

$= 1200 \times 3.14 \quad [\because \pi = 3.14]$

$= 3768 \text{ cm}^3$

122. *(d)* $\because$ Surface area of cone $ABD = \pi rl$

$= \pi \times 12 \times 15 = 180\pi \text{ cm}^2$

and surface area of cone

$BCD = \pi \times 12 \times 20 = 240\pi \text{ cm}^2$

$\therefore$ Required surface area of double cone

$= 180\pi + 240\pi = 420\pi$

$= 420 \times 3.14 = 1318.8 \text{ cm}^2$

123. *(c)* Let side of a cube be x.
Then, surface area of cube $= 6x^2$
If the side of cube is increased by 100%.

Then, new side of cube $= x + 100\%$ of x

$= x + x = 2x$

$\therefore$ New surface area of cube $= 6(2x)^2$

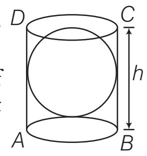
$= 6 \times 4x^2 = 24x^2$

Now, increase percentage in surface area

$= \frac{24x^2 - 6x^2}{6x^2} \times 100$

$= \frac{18}{6} \times 100 = 300\%$

124. *(a)* Let radius of the sphere be r.



Since, cylinder circumscribes a sphere.

$\therefore$ Radius of the base of cylinder $= r$ and height of cylinder $= 2r$

$=$ Diameter of sphere

Now, volume of sphere $= \frac{4}{3}\pi r^3$

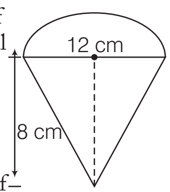
and volume of cylinder

$= \pi r^2 h = \pi r^2 (2r) = 2\pi r^3$

$\therefore$ Required ratio $= \frac{\frac{4}{3}\pi r^3}{2\pi r^3} = \frac{4}{3 \times 2}$

$= \frac{2}{3} = 2 : 3$

125. *(a)* Given, diameter of the base of the conical portion $= 12$ cm



$\therefore$ Radius of conical portion $= 6$ cm

Radius of hemisphere $= 6$ cm and height of conical portion $= 8$ cm

$\therefore$ Slant height of conical portion

$= \sqrt{6^2 + 8^2} = \sqrt{36 + 64}$

$[\because l = \sqrt{r^2 + h^2}]$

$= \sqrt{100} = 10$ cm

Now, total surface area of the toy

$=$ Curved surface area of conical portion + Curved surface area of hemisphere

$= \pi rl + 2\pi r^2 = \pi(rl + 2r^2)$

$= \pi(6 \times 10 + 2 \times 6 \times 6)$

$= \pi(60 + 72) = 132\pi \text{ cm}^2$

126. *(b)* Volume of the toy

$=$ Volume of conical portion + Volume of hemisphere

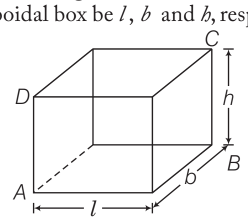
$= \frac{1}{3}\pi r^2 h + \frac{2}{3}\pi r^3$

$= \pi\left(\frac{1}{3}r^2 h + \frac{2}{3}r^3\right)$

$= \pi\left[\frac{1}{3} \times (6)^2 \times 8 + \frac{2}{3} \times (6)^3\right]$

$= \pi[96 + 144] = 240\pi \text{ cm}^3$

127. *(b)* Let length, breadth and height of a cuboidal box be l, b and h, respectively.



Given, areas of the three adjacent faces are x, $4x$ and $9x$ sq units.

Now, $lb = x$

$[\because$ area of rectangular face $=$ length $\times$ breadth$]$

Similarly, $bh = 4x$ and $lh = 9x$

Now, $(lb) \cdot (bh) \cdot (lh) = (x) \cdot (4x) \cdot (9x)$

$\Rightarrow \quad (lbh)^2 = 36x^3 \Rightarrow lbh = \sqrt{36x^3}$

$\therefore \quad lbh = 6x^{3/2}$

Hence, volume of cuboidal box

$= lbh = 6x^{3/2}$ cu units

128. *(c)* In a cuboid, 4 perpendicular face pairs in bottom surface, 4 perpendicular face pairs in top surface and 4 perpendicular face pairs in vertical surface.

Hence, total perpendicular pairs are 12.

129. *(c)* I. Surface area of sphere A

$= 4\pi(6)^2 = 144\pi \text{ cm}^2$

Surface area of sphere $B = 4\pi(8)^2$

$= 256\pi \text{ cm}^2$

Surface area of sphere $C = 4\pi(10)^2$

$= 400\pi \text{ cm}^2$

Now, sum of surface area of spheres A and B

$= 144\pi + 256\pi = 400\pi \text{ cm}^2$

$=$ Surface area of sphere C

Hence, statement I is correct.

II. $\because$ Volume of sphere D

$= \frac{4}{3}\pi(12)^3 = 2304\pi \text{ cm}^3$

Volume of sphere $A = \frac{4}{3}\pi(6)^3 \text{ cm}^3$

Volume of sphere $B = \frac{4}{3}\pi(8)^3 \text{ cm}^3$

and volume of sphere

$C = \frac{4}{3}\pi(10)^3 \text{ cm}^3$

Now, sum of volumes of spheres A, B and C

$= \frac{4}{3}\pi[6^3 + 8^3 + 10^3]$

$= \frac{4}{3}\pi[216 + 512 + 1000]$

$= 2304\pi \text{ cm}^3 =$ Volume of sphere D

Hence, statement II is also correct.